

RESEARCH AND EDUCATION

Dual convolutional neural network framework for segmenting dental caries in panoramic radiographs

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Dental caries is one of the most prevalent chronic diseases affecting people of all ages worldwide.¹ According to the American Dental Association, dental caries can be classified into 4 stages based on the extent of the lesion: sound, initial, moderate, and extensive.^{2,3} Timely diagnosis of carious lesions is crucial for preventing disease progression, reducing the need for invasive treatments, and ultimately lowering healthcare costs.^{4,5} However, because of the complexity of dental anatomy and the location of lesions, accurate diagnosis can be challenging, especially for proximal lesions, which are difficult to detect through visual inspection or tactile examination alone.^{6–8}

ABSTRACT

Statement of problem. Dental caries, a widespread chronic disease, has been difficult to detect, especially in posterior proximal regions. Conventional diagnostic methods, such as visual inspection and radiography, are subjective and inconsistent across clinicians.

Purpose. The purpose of this study was to develop and evaluate a deep learning-based method for automated detection and segmentation of dental caries in panoramic radiographs.

Material and methods. A deep learning pipeline combining Faster Regions based Convolutional Neural Networks (R-CNN) and U-Net architectures was developed. The Faster R-CNN model was used to detect tooth regions, and the U-Net model segmented carious areas within these regions. Performance differences against comparative models were evaluated for statistical significance using paired *t* tests ($\alpha=.05$).

Results. The proposed method achieved an intersection over union of 0.6075, a dice coefficient of 0.7429, a recall of 0.7309, and a precision of 0.7881. This performance represented an improvement in intersection over union, dice coefficient, and recall over conventional segmentation models, with the difference being statistically significant ($P<.05$).

Conclusions. The results indicated that the proposed deep learning method was effective in detecting and segmenting dental caries in panoramic radiographs and showed potential for improving diagnostic accuracy. (J Prosthet Dent xxxx;xxx:xxx-xxx)

To address these limitations, various alternative diagnostic technologies such as fiber optic transillumination and digital fluorescence-based systems (DIAGNOdent;

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Clinical Implications

The integration of a deep learning-based model for caries detection and segmentation in panoramic radiographs may assist clinicians by providing auxiliary diagnostic information, particularly in complex or ambiguous situations. By supporting consistent interpretation of radiographic findings, such tools have the potential to reduce diagnostic variability and improve workflow efficiency in dental practice.

KaVo) have been proposed. These methods assist in detecting carious lesions by sensing changes in the optical properties of enamel and dentin.⁹ However, their diagnostic performance can vary depending on clinical settings, and they require expensive, specialized equipment, which limits their widespread application.^{10,11} As a result, among radiographic techniques, bitewing radiography has become a standard tool in dental diagnostics.^{12,13} In particular, bitewing radiographs are effective in visualizing the proximal surfaces where caries most commonly occurs, providing critical diagnostic information for lesions that are difficult to detect through visual inspection alone. Consequently, the combined use of visual examination and radiographic imaging has become a widely adopted clinical approach for the routine diagnosis of proximal caries.^{14,15} However, bitewing radiographs have a limited field of view, making it difficult to assess the overall oral condition or the distribution of lesions across multiple teeth. To address these limitations, panoramic radiographs are sometimes used.¹⁶ Panoramic radiographs allow visualization of the entire maxillary and mandibular dentition, along with surrounding structures, in a single exposure, making it useful for identifying the overall distribution of lesions. However, panoramic radiographs have inherent limitations, such as image distortion, reduced resolution, and structural overlap.^{17–20}

In addition to these technical limitations, radiographic interpretation is also subject to considerable variability among clinicians. Factors such as image quality, viewing conditions, clinical experience, and individual diagnostic tendencies can all influence diagnostic outcomes. Wenzel et al²¹ reported kappa values ranging from 0.30 to 0.72 among 34 raters who interpreted the same set of bitewing radiographs, indicating substantial interrater variability. This subjectivity hampers the reliable evaluation of carious lesions and underscores the need for more standardized and consistent diagnostic systems.

The rise of artificial intelligence (AI) and deep learning has opened new possibilities for automating and enhancing diagnostic accuracy in various medical

fields, including dentistry.^{22–25} Convolutional neural network (CNN), a class of deep learning models designed to process visual data, have shown great promise in analyzing medical images with improved precision and speed.^{22,26,27} In dentistry, CNN have been applied to tasks such as diagnosing caries, identifying fractures, assessing periodontal bone loss, and detecting periapical lesions from dental radiographs.^{28–30} Among the pioneering deep learning architectures, the U-Net model, introduced by Ronneberger et al,³¹ has demonstrated effectiveness in medical image segmentation and analysis, particularly for dental structure segmentation in bitewing radiographs. The U-Net architecture's ability to capture both local and global contextual information makes it highly suitable for identifying dental caries, which often manifest as subtle variations in radiographic density. Subsequent studies have further explored deep learning models for dental caries detection, comparing their diagnostic performance with that of clinicians. For instance, Casalegno et al³² reported that a CNN-based model achieved comparable diagnostic accuracy to expert clinicians in detecting dental caries.

The aim of the present study was to address the distortion and resolution limitations of panoramic radiographs using deep learning to enhance the accuracy of carious lesion segmentation. Notably, few studies have focused on tooth-level segmentation of carious lesions across entire panoramic images. A fully automated diagnostic framework was proposed, based on panoramic radiographs integrated Faster R-CNN and U-Net, enabling the detection of individual teeth followed by the automatic segmentation of carious lesions for each tooth. This approach allowed for effective identification and delineation of carious regions and automatically highlighted suspected lesion areas. The null hypothesis was that there would be no statistically significant performance difference between the proposed method and the comparative segmentation models.

MATERIAL AND METHODS

This study used an open-source dataset composed of 597 panoramic radiographs collected from the Department of Stomatology at Zhejiang Provincial People's Hospital, with labels annotated by 5 experienced dentists to ensure accuracy and consistency.³³ The dataset consisted of 497 training images and 100 testing images. From the training set, 4 images without caries annotations were removed, leaving 493 panoramic images. These were first split into 443 for training and 50 for validation without stratification at the panoramic image level, after which the teeth were cropped and class balance in the training set was addressed at the tooth level by augmentation.³⁴ The proposed method for

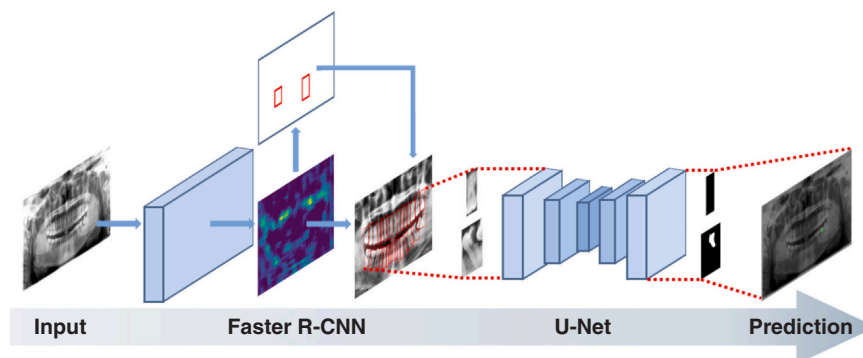


Figure 1. Overall process of proposed method for dental caries segmentation.

dental caries segmentation followed a 2-step process, as illustrated in Figure 1.

In the first stage, several preprocessing steps were applied to accurately detect individual teeth. Since panoramic radiographs include complex facial structures outside the oral region that can negatively impact model performance, the images were cropped to focus solely on the oral region, retaining 20% to 80% of the image width and 20% to 90% of the height. Subsequently, Contrast Limited Adaptive Histogram Equalization (CLAHE) was applied to enhance the contrast and improve the clarity of the radiographic images.³⁵ Based on the preprocessed images, a Faster R-CNN model, a widely used convolutional neural network architecture for object detection, was used to identify the locations of teeth.³⁶ In the second stage, a U-Net model was employed for binary segmentation of dental caries within the detected tooth regions.³¹ The Faster R-CNN was trained on 443 panoramic radiographs and validated on 50 images. U-Net training and validation datasets were

made by cropping individual tooth images from the Faster R-CNN train and validation sets. To address class imbalance, augmentation increased the 3786 carious images fourfold to 15 144 via horizontal, vertical, and combined flips, while the 9304 noncarious images were doubled to 18 608 using vertical flips only. U-Net was trained on the augmented set of 33 752 images.³⁷ The validation set consisted of 1477 images, including 224 with caries. Both models leveraged pre-trained backbone networks. Finally, the predicted lesion masks were mapped onto the original panoramic radiographs for visual verification by clinicians.

The Faster R-CNN model, used in the first stage of tooth detection, was an established object detection framework that combined a Region Proposal Network for generating candidate object regions and a Fast R-CNN classifier for object classification and bounding box refinement (Fig. 2).³⁸ To increase computational efficiency, MobilenetV2 was used as the backbone, with the advantage of reducing computational overhead while

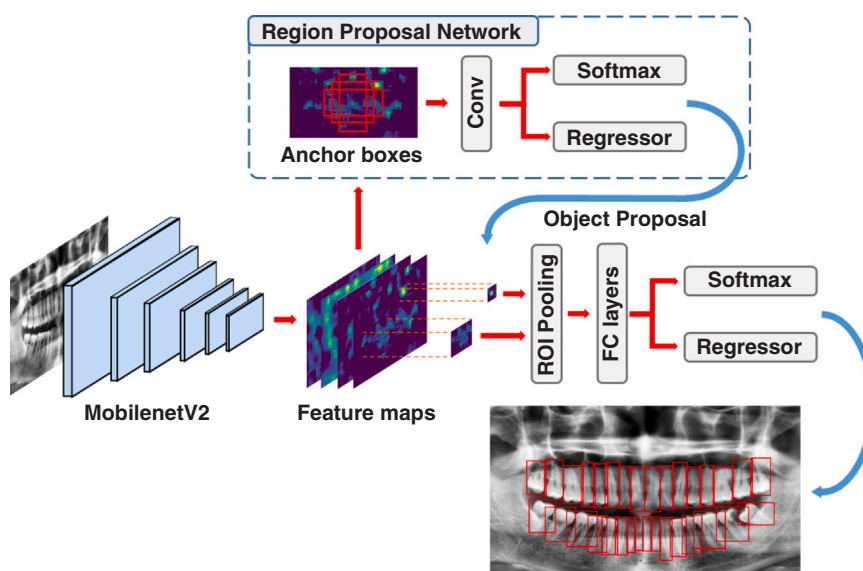


Figure 2. Architecture of Faster R-CNN with MobilenetV2 as backbone, illustrating example of tooth detection and result of cropping image using bounding box coordinates.

maintaining accuracy through the use of depth wise separable convolutions and an Inverted Residual structure.³⁹ Training efficiency was improved with transfer learning using pretrained weights from the ImageNet dataset, and the model's final output was the bounding box coordinates for each detected tooth. The U-Net model, employed for the second stage of dental caries segmentation, featured a U-shaped architecture designed for semantic segmentation. It consisted of a contracting path that extracted features while compressing the image and an expansive path that generated pixel-level segmentation by upscaling the feature maps.^{31,40} For the present study, EfficientNet-B0 was chosen as the backbone for U-Net, considering the balance between computational efficiency and performance.⁴¹ Figure 3 shows the overall U-Net structure incorporating EfficientNet-B0 as the encoder. For input to the U-Net, the cropped tooth images obtained from the bounding boxes output by Faster R-CNN were resized to 256×256 pixels.

The training and testing of the proposed model were conducted in an environment (Python 3.10.13; Python Software Foundation) using a library (Pytorch 2.1.2; PyTorch Foundation). All experiments were carried out in an online notebook environment (Kaggle Notebooks; Kaggle), which provided a central processing unit (Xeon; Intel), 29 GB random access memory, and a graphics processing unit (Tesla P100; NVIDIA) for model training and evaluation. To optimize the performance of both the Faster R-CNN and U-Net models, key hyperparameters such as batch size, optimizer, learning rate, and loss function were fine-tuned empirically. The total loss function of the Faster R-CNN model was a composite loss that simultaneously handled object detection and bounding box regression. It included both classification

and regression components, allowing the model to learn how to identify objects and accurately adjust their positions.

Because of the imbalance between the carious and unaffected regions in the tooth images, Dice Loss was selected as the loss function for U-Net. Dice Loss calculated the loss based on the Dice coefficient, which was defined as $\text{Dice} = (2 \times \text{TP}) / (2 \times \text{TP} + \text{FP} + \text{FN})$, where TP represents True Positive, FP represents False Positive, and FN represents False Negative. Dice Loss was defined as 1 minus the value of the Dice coefficient. The optimizer used for Faster R-CNN was Adam with a learning rate of 0.001, while AdamW with a learning rate of 0.0001 and a weight decay coefficient of 0.001 was chosen for U-Net to enhance the generalization performance of the Adam algorithm.^{42,43} Batch sizes were set to 4 for Faster R-CNN and 32 for U-Net, and training was conducted for 100 and 50 epochs, respectively. Model performance was evaluated at each epoch by calculating the Intersection over Union (IoU) between predicted and ground truth regions, and the model achieving the highest IoU was saved.⁴⁴ The final performance of the proposed method was evaluated using 4 key metrics. The widely used IoU measured the overlap between the predicted segmentation mask and the ground truth mask and was calculated as $\text{TP} / (\text{TP} + \text{FP} + \text{FN})$. Recall, calculated as $\text{TP} / (\text{TP} + \text{FN})$, evaluated how well the model identified true positive instances. Precision, calculated as $\text{TP} / (\text{TP} + \text{FP})$, measured the accuracy of the model's positive predictions. The Dice coefficient, previously used as a loss function, calculated the overlap between the predicted and actual positive regions and provided a balanced measure of segmentation performance. To determine whether the performance differences between the proposed model and the comparative models were

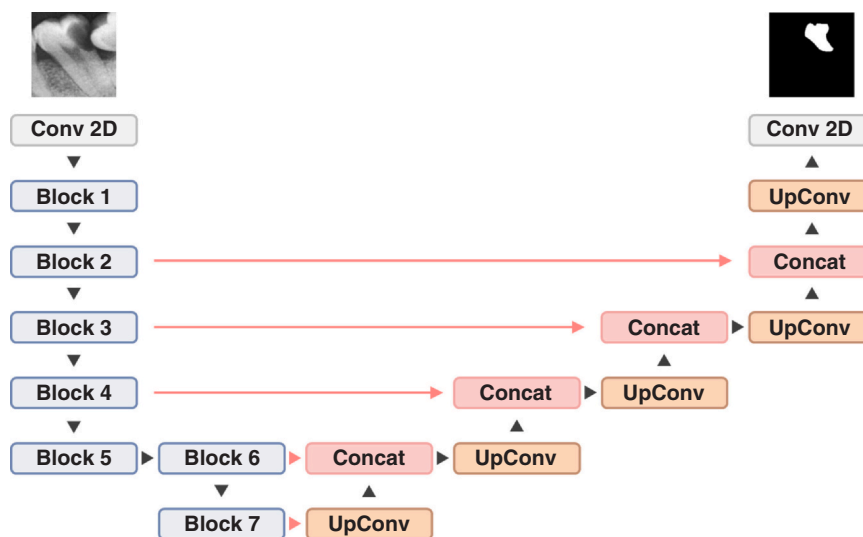


Figure 3. Schematic of U-Net architecture with EfficientNet-B0 encoder. Encoder used initial EfficientNet-B0 blocks, beginning with mobile inverted bottleneck convolution (MBConv) stage MBConv1, followed by successive MBConv6 stages to extract multiscale features.

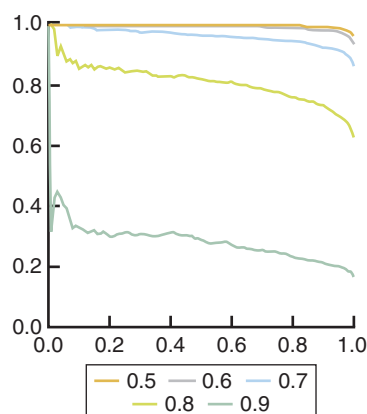


Figure 4. Precision-Recall curve evaluating tooth detection performance of Faster R-CNN model at IoU thresholds ranging from 0.5 to 0.9.

statistically meaningful, a paired t test was conducted for each metric ($\alpha=.05$).

RESULTS

Figure 4 shows the precision–recall curves computed by stepwise variation of the IoU threshold between the model's predictions and the ground truth tooth annotations from 0.5 to 0.9, confirming that detection performance gradually declined as the IoU threshold increased. At IoU thresholds between 0.5 and 0.8, the model maintained high precision and recall and exhibited stable detection performance. The corresponding Average Precision (AP) scores across the IoU thresholds further substantiated the model's performance: 0.9981 at IoU 0.5, 0.9940 at 0.6, and 0.9635 at 0.7 and reflected consistently high detection accuracy. An IoU of 0.80 was generally regarded as a stricter localization threshold.^{45,46} Under this criterion, the model achieved an AP of 0.8124. However, at the more stringent threshold of 0.9, the AP markedly declined to 0.2819 and suggested a notable degradation in performance as the localization criterion became increasingly strict.

Figure 5 shows a magnified view of the oral region in the panoramic image. The first column show the original panoramic image of the oral region, while the second and third columns show the ground truth and predicted

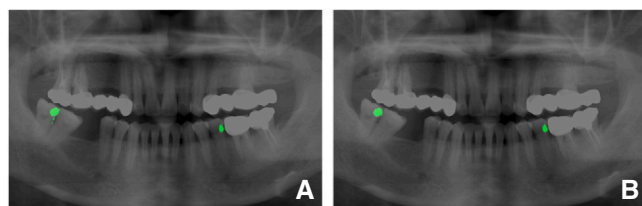


Figure 5. Caries segmentation results in panoramic radiographic images. A, Ground Truth. B, Predicted.

segmentation, respectively. The model's predictions for carious regions were displayed as green overlays. By magnifying the oral region, the comparison between the predicted segmentation and the ground truth was more evident. The model successfully identified the carious regions in the panoramic image, accurately highlighted the affected areas. The overall performance of the model was evaluated on the test dataset using several metrics. The method achieved an average IoU of 0.608 (95% confidence interval [CI]: 0.577–0.638), which measured the overlap between the predicted and actual carious regions. The Dice coefficient, which reflected the similarity between the ground truth and the predictions, was 0.743 (95% CI: 0.716–0.770). The Recall value was 0.731 (95% CI: 0.697–0.765). The precision was 0.788 (95% CI: 0.762–0.815). In contrast, the recall was 0.7309, suggesting the possibility that certain lesions were not detected. While both precision and recall demonstrated moderate to high levels of performance, further enhancement in detecting smaller or less well-defined lesions may require additional data acquisition and architectural improvements to the model.

To validate the performance of the proposed method, a comparative experiment was conducted with widely used deep learning-based networks for medical image segmentation, including U-Net, U-Net++, Multi-Scale Attention Network (MA-Net), and DeepLab V3+.^{47–49} Each network was trained using full panoramic images as input and under the same dataset configuration as the proposed method. The performance was compared based on the average values of 4 evaluation metrics including IoU, Dice coefficient, Recall and Precision. All models were trained using the same hyperparameter settings for batch size, input image size, and learning rate, and EfficientNet-B0 was used as the backbone for all networks. The batch size was set to 16, and the input images were resized to 512×512 pixels. Training was conducted over a total of 50 epochs. The optimizer used was AdamW, and the learning rate was fixed at 0.0001 and applied consistently across all experiments. To enhance the diversity of the training data, horizontal flip-based data augmentation was applied. As a result, a total of 886 training images were obtained from the original 443 images.

As shown in Table 1, the proposed method outperformed the comparison models in terms of IoU, Dice, and Recall. A paired t test confirmed that these improvements were statistically significant ($P<.05$). In contrast, the precision score did not show a significant difference when compared to U-Net or U-Net+ ($P>.05$).

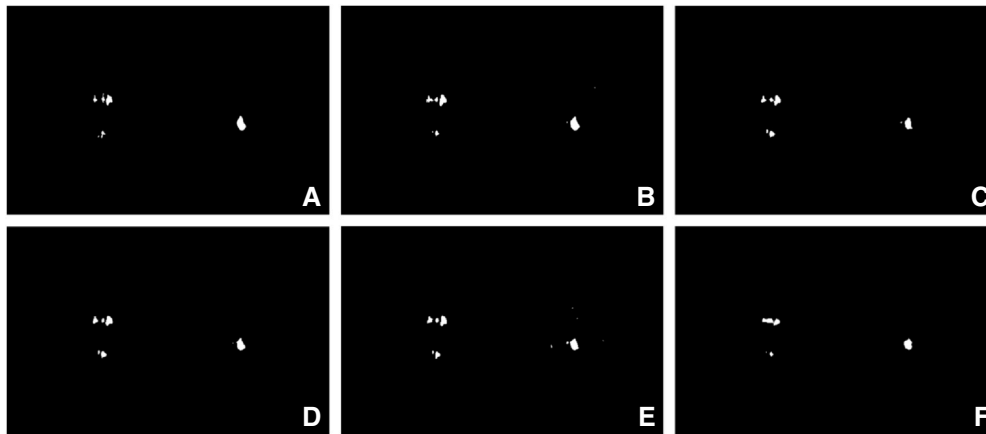
Figure 6 presents a qualitative comparison of segmentation results on representative test samples. The proposed method demonstrated a high degree of morphological agreement with the ground truth, accurately

Table 1. Comparison of Segmentation Performance on the Test Dataset Using Different Deep Learning Models (n=100)

Model	IoU (95% CI)	Dice (95% CI)	Recall (95% CI)	Precision (95% CI)
U-Net	0.5322 (0.5011, 0.5634)	0.6790 (0.6486, 0.7094)	0.6329 (0.5937, 0.6722)	0.7850 (0.7558, 0.8143)
U-Net++	0.5426 (0.5143, 0.5709)	0.6909 (0.6641, 0.7178)	0.6545 (0.6177, 0.6913)	0.7789 (0.7517, 0.8064)
MA-Net	0.5287 (0.4982, 0.5589)	0.6763 (0.6456, 0.7067)	0.6496 (0.6118, 0.6873)	0.7454 (0.7131, 0.7774)
DeepLab V3+	0.4593 (0.4279, 0.4906)	0.6104 (0.5755, 0.6452)	0.5889 (0.5473, 0.6302)	0.6962 (0.6602, 0.7321)

CI, confidence interval; Dice, Dice coefficient; IoU, Intersection of Union

Each model was trained with the same configuration, and the results were evaluated in terms of IoU, Dice coefficient, Recall, and Precision.

**Figure 6.** Proposed method and comparative model results. A, Ground Truth. B, Proposed method. C, U-Net. D, U-Net++. E, MA-Net. F, DeepLabV3+.

delineating the boundaries of carious lesions. In comparison, the baseline models frequently failed to detect small lesion areas or produced fragmented predictions. These findings indicated that the proposed method could capture the fine structural details of carious regions and achieving more precise and complete segmentation outcomes.

DISCUSSION

The proposed method achieved statistically significant gains over the baseline models in IoU, Dice, and recall ($P < .05$), whereas the differences in precision were not significant against certain baselines ($P > .05$). Consequently, the null hypothesis that there would be no statistically significant performance difference between the proposed method and the comparative segmentation models was rejected for IoU, Dice, and recall, but not for precision. The proposed method demonstrated high performance on both tooth detection and caries segmentation tasks using only 2D panoramic radiographs. In the tooth detection task, precision and recall both remained above 0.99 for IoU thresholds from 0.5 to 0.8, and AP was measured at 0.9981 for IoU 0.5, 0.9940 for IoU 0.6, 0.9635 for IoU 0.7, and 0.8124 for IoU 0.8, confirming stable localization of teeth in panoramic images. However, under the stringent IoU 0.9 criterion, AP fell to 0.28. This performance degradation appeared to be attributable to the inherent characteristics of panoramic radiographs, such as geometric distortion and

the overlapping of anatomical structures. Clinically, this limitation could affect the precise detection of small or early-stage carious lesions. Therefore, future work should involve additional image preprocessing or improvements in the model architecture to compensate for these factors. For caries segmentation, the proposed method achieved an IoU of 0.61, a Dice coefficient of 0.74, a recall of 0.73, and a precision of 0.79. However, the clinical significance of this performance is unclear without a human benchmark comparison. The recall of 0.73 was particularly concerning, as it indicated that the system missed nearly 27% of carious lesions. This high risk of false negatives, which could delay treatment, represented a critical limitation for clinical application. Therefore, future work must prioritize improving recall and validating clinical viability against a human expert benchmark. While medium to large lesions were segmented reliably, small or poorly defined lesions still experienced some missed detections, an outcome likely attributable to having trained a single segmentation model without accounting for variations in X-ray transmission and absorption characteristics according to tooth thickness. Furthermore, the limited size of the dataset, with only 100 images available for testing, was a key limitation. Although data augmentation increased the diversity of the training data, it could not completely replace the broad clinical variability. Therefore, external validation is essential to verify the robustness of the results.

The proposed method first detected teeth in panoramic dental radiographs and then performed

segmentation on the detected regions, which reduced spurious learning of background patterns. Compared with single stage approaches, it yielded statistically significant improvements in lesion segmentation metrics, including IoU, Dice, and recall, across all comparison models. Precision did not differ significantly from U-Net and U-Net++, but it exceeded that of the other comparison models, and recall was improved without compromising precision. Although an IoU of 0.61 and a Dice of 0.74 were moderate in general computer vision settings, they represented a statistically significant improvement over the comparison models for dental caries segmentation on panoramic radiographs.

Future research should introduce tooth type classification, including incisors, canines, premolars, and molars, during the detection stage and then train and predict with separate segmentation models for each tooth type. Expanding annotations to reflect tooth specific radiographic translucency and adding multiclass labels for varying translucency will enable the model to learn subtle structural differences. Incorporating multi-stage annotations based on caries depth will allow precise representation of lesion severity. Finally, validating generalization through cross-validation on external panoramic radiograph datasets and conducting clinical deployment studies to assess usability, diagnostic efficiency, and patient outcomes will be essential. These efforts are expected to translate the findings into a practical clinical support tool.

CONCLUSIONS

Based on the research findings, the following conclusions were drawn:

1. The proposed deep learning-based framework demonstrated effective detection and segmentation of dental caries in panoramic radiographs.
2. The proposed method, which used detected teeth as input, achieved statistically significant improvements in IoU, Dice coefficient, and recall on the test dataset compared with U-Net, U-Net++, MA-Net, and DeepLab V3+, which used panoramic radiographs.
3. The limitations in detecting subtle lesions suggested the necessity of enhancing performance in future studies by expanding the dataset and refining the model architecture.

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